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3. (a) Give a detailed account of the major lymphoid organs and their distribution in fishes. 8
(b) Discuss the evolution of Toll-like receptors from *Drosophila* Toll-1 to mammalian TLRs. 7
4. (a) Give a comparative account of peripheral lymphoid organs in birds and other vertebrates with respect to their evolution and significance 10
(b) Describe the role of head kidney (pronephros) as a primary lymphoid organ in fishes 5
5. (a) Describe the evolutionary diversification of cytokines in vertebrates with reference to their functional significance across vertebrate classes. 9
(b) Explain the process of immunological memory development in vertebrates. How do primary and secondary immune responses differ from each other? 6
6. Write a short note on (attempt any **three**): 15
(i) Variable Lymphocyte Receptors (VLRs)
(ii) Evolution of MHC in teleosts
(iii) Evolution of immunoglobulins
(iv) Evolution of immunotolerance

[This question paper contains 4 printed pages]

Your Roll No.

: 18.05.2026(E)

Sl. No. of Q. Paper

: 7470

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Unique Paper Code

: 2233520002

Name of the Paper

: Evolutionary Immunobiology

Name of the Course

: B.Sc. (H) Zoology (NEP)

Semester

: VIII

Time : 3 Hours

Maximum Marks : 90

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
(b) Attempt any **FIVE** questions in **all**.
(c) Question No. 1 is compulsory.

1. (a) Define:

1×6=6

- (i) Opsonization
(ii) Variable Lymphocyte Receptors
(iii) Somatic Hypermutation
(iv) Thymic Selection
(v) Cytokine Pleiotropy
(vi) Antimicrobial peptide

(b) Write exact location and function of the following : $1 \times 6 = 6$

- (i) Hassall's corpuscles
- (ii) Harderian gland
- (iii) Peyer's Patches
- (iv) GALT
- (v) Leydig Organ
- (vi) Mural nodules

(c) Differentiate Between (Any **three**) $2 \times 3 = 6$

- (i) Toll-1 Receptors in *Drosophila* and Toll-Like Receptors in Mammals
- (ii) Thymus of birds and mammals
- (iii) Mammalian and teleost MHC organization
- (iv) Paralogous and orthologous gene duplication

(d) State **True** or **False** : $1 \times 6 = 6$

- (i) All invertebrates, possess hemocytes that perform phagocytosis as part of their innate defence mechanism.
- (ii) Bacteria possess immune factors which provide defence against viral pathogen
- (iii) Pro-inflammatory cytokines (IL- 1β ; TNF- α) are exclusive to mammals and have no functional homologs in lower vertebrates.

(iv) The Toll-1 receptor in *Drosophila* directly recognises and binds to fungal cell wall components to initiate the innate immune response.

(v) Jawless vertebrates completely lack adaptive immunity and rely solely on innate immune mechanisms for host defence.

(vi) IL-10 is an anti-inflammatory cytokine that suppresses pro-inflammatory immune responses

(vii) Both the alternative and lectin complement-activation pathways are present in ascidians

(e) Expand the following : $1 \times 6 = 6$

- (i) PRRs
- (ii) RAG
- (iii) AMP
- (iv) MMCs
- (v) LRR
- (vi) AID

2. (a) Compare and contrast the development of lymphocytes in different vertebrate classes. How do these variations reflect the evolutionary refinement of adaptive immunity? 8

(b) Discuss the evolution of cytokines across various classes of vertebrates. 7